Central Question:

- Do regions with higher pesticide use have higher contamination concentrations in surface water?
    - Variables
        - Pesticide name, estimated (high) pesticide use, county fips, data collection year, pesticide water concentration
    - Observations
        - Our initial analysis compared pesticide use (by compound) to pesticide contamination
        - This relationship did not reveal a strong correlation
    - Intended comparisons
        - To further refine this
            - We are going to play around with log scaling the axes to see if this helps reveal a relationship
            - We are going to categorize the pesticides (herbicide, fungicide, insecticide, etc.) to see if there is greater correlation for specific compounds.
            - We are going to look at total pesticide use and total pesticide contamination (all compounds)
    - Limitations
        - Uncertainty in the relationship between use and contamination within a county.
        - Without insight into watershed behavior, we cannot be certain that pesticides that are used within one county will contaminate the waters within that same county.

Supporting Questions:

- Are there stronger correlations of pesticide use and contamination among different pesticide categories (herbicides, fungicides, insecticides, etc.)?
    - Variables
        - Pesticide category, estimated (high) pesticide use, county fips, data collection year, pesticide water concentration
    - Intended comparisons
        - Compare pesticide category use by pesticide category contamination
    - Limitations
        - We need to figure out how to group our current pesticide data into their respective categories.
- How has pesticide use and average pesticide contamination changed over time?
    - Variables

- Pesticide category, estimated (high) pesticide use, county fips, data collection year, pesticide water concentration
- Intended comparisons
    - Compare total pesticide use for each year (2013-2017)
    - Compare average contamination concentration for each year (2013-2017)